

NETWORK TYPES

UNDERSTANDING DIFFERENT TYPES OF NETWORKS
AND HOW THEY CONNECT THE WORLD.



WAN



LAN / WLAN



VPN



MAN / GAN



PAN / WPAN

From global connections to personal networks –
explore the types, topologies, and technologies
that keep us **connected**.



02

NETWORK TYPES

AN OVERVIEW

Networks come in different types and topologies, designed for various needs and environments. This slide gives a high-level overview of the main categories we'll cover.



WAN

Connects networks across large geographical areas.



VPN

Creates secure tunnels to connect remote users or whole networks.



LAN / WLAN

Connects devices within a local area using cables or Wi-Fi.



OTHER (Book Terms)

Includes GAN, MAN, PAN / WPAN – used in theory and literature.



03 WIDE AREA NETWORK (WAN)

The WAN connects multiple LANs across large geographical areas. The Internet is the most common example of a WAN.



WHAT IS IT?

A network that spans large geographical areas, connecting multiple LANs together.



IDENTIFICATION

Uses WAN-specific routing protocols (e.g., BGP) and non-RFC1918 IP address ranges.



ADDRESSING

Uses public IP addresses that are routable on the Internet.



EXAMPLES

The Internet, corporate WANs, government networks, intranets.



04 VIRTUAL PRIVATE NETWORK (VPN)

A VPN creates a secure, encrypted connection over the Internet, allowing users or entire networks to communicate as if they were on the same local network.

TYPES OF VPN



SITE-TO-SITE VPN

Connects entire networks between two or more locations using secure encrypted tunnels.



REMOTE ACCESS VPN

Allows individual users to securely connect to a remote network from anywhere.



SSL VPN

Provides secure access to applications or desktops through a web browser.

HOW VPN WORKS



SECURE

Encrypts data to protect against eavesdropping.



PRIVATE

Hides your IP address and location.



ACCESS

Access internal resources securely.



RELIABLE

Ensures data integrity and secure connections.



FLEXIBLE

Supports remote work and multi-site connectivity.

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LOCAL AREA NETWORK (LAN) / WIRELESS LAN (WLAN)

LANs and WLANs connect devices within a limited area such as a home, office, or building. The main difference is the use of cables (LAN) or Wi-Fi (WLAN).

KEY POINTS



IP ADDRESSING

Typically uses private IP ranges (RFC 1918):
10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.



PURPOSE

Designed for local connectivity, resource sharing,
and internal communication.



WLAN BENEFIT

Provides mobility and flexibility by allowing devices
to connect wirelessly.



SECURITY FOCUS

Security is the main consideration when
implementing WLAN.



LAN (WIRED)

- Uses Ethernet cables
- More stable and faster
- Ideal for fixed devices
- Higher physical security

VS



WLAN (WIRELESS)

- Uses Wi-Fi for connectivity
- More flexible and mobile
- Easier to deploy
- Requires stronger security

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VIRTUAL PRIVATE NETWORK (VPN)

VPNs extend private networks over the Internet securely. Different types of VPNs achieve the same goal – making users feel like they are on a different network as if they were plugged in locally.

KEY BENEFITS



SECURE CONNECTION

Encrypts data to protect against eavesdropping and unauthorized access.



REMOTE ACCESS

Access resources and networks from anywhere in the world.



COST EFFECTIVE

Reduces the need for expensive leased lines between locations.



FLEXIBILITY

Supports multiple use cases and deployment scenarios.

THREE MAIN TYPES OF VPN

1 SITE-TO-SITE VPN

Connects entire networks between two or more locations using encrypted tunnels.



COMMON USE



Connecting branch offices, data centers, or cloud networks securely.

2 REMOTE ACCESS VPN

Allows individual users to securely connect to a private network from anywhere.



COMMON USE



Employees accessing company resources from home, while traveling, or on the go.

3 SSL VPN

Provides VPN access through a web browser using SSL/TLS encryption.



COMMON USE



Browser-based access to applications or desktops without client software.



All VPN types use **encryption** to protect data. The choice depends on your needs, resources, and the level of access required.



07 BOOK TERMS

(THEORY & LITERATURE)

These terms are commonly used in literature and examinations. They categorize networks based on geographical range and coverage.



WHY THESE TERMS MATTER

Understanding these classifications helps in designing, scaling, and managing networks across personal, local, regional, and global levels.



CLASSIFY

Categorize networks by coverage area.



UNDERSTAND

Know the scope and purpose.



DESIGN

Choose the right network for the need.



SECURE

Implement security at every level.

NETWORK TYPES IN THEORY

01 GLOBAL AREA NETWORK (GAN)

A worldwide network spanning continents and countries.



EXAMPLES

- The Internet
- International corporate networks
- Global data exchange systems

02 METROPOLITAN AREA NETWORK (MAN)

Connects multiple LANs within a city or region.



EXAMPLES

- City-wide networks
- Multiple branches linked via leased lines
- ISP metropolitan networks

03 PERSONAL AREA NETWORK (PAN)

A short-range network around a person, usually via cable.



EXAMPLES

- USB-connected devices
- Tethering via cable
- Personal device connections

04 WIRELESS PERSONAL AREA NETWORK (WPAN)

Short-range wireless network using Bluetooth or similar technologies.



EXAMPLES

- Bluetooth (Piconet)
- Wireless USB
- IoT devices with low data transmission



From personal connections to the global Internet – networks connect everything.



Choose the right network type for **efficiency**, **performance**, and **security**.



08 OSI MODEL

(OPEN SYSTEMS INTERCONNECTION)

The OSI Model is a conceptual framework that standardizes how data is transmitted across a network. It divides network communication into 7 distinct layers.



WHY OSI MODEL MATTERS?

- Helps in understanding network functions clearly.
- Simplifies troubleshooting.
- Ensures vendor-independent communication.
- Foundation for modern networking protocols.



STANDARDIZED

Provides a common language for networking.



MODULAR

Each layer has a specific role and responsibility.



EASY TROUBLESHOOTING

Issues can be isolated to a specific layer.



FLEXIBLE

Changes in one layer don't affect others.

THE 7 LAYERS OF OSI MODEL

7		APPLICATION LAYER	Provides network services directly to user applications. Examples: HTTP, FTP, SMTP, DNS
6		PRESENTATION LAYER	Handles data representation, encryption, and compression. Examples: SSL/TLS, JPEG, ASCII
5		SESSION LAYER	Establishes, manages, and terminates sessions between applications. Examples: NetBIOS, RPC
4		TRANSPORT LAYER	Ensures end-to-end delivery, error control, and flow control. Examples: TCP, UDP
3		NETWORK LAYER	Handles logical addressing and routing of data packets. Examples: IP, ICMP, OSPF
2		DATA LINK LAYER	Provides node-to-node data transfer and error detection. Examples: Ethernet, MAC, PPP
1		PHYSICAL LAYER	Transmits raw bits over the physical medium. Examples: Cables, Hubs, Signals

HOW DATA TRAVELS



KEY TAKEAWAY

Each layer communicates with the one above and below it, using standardized protocols to ensure smooth and reliable data transmission.

09 NETWORK SECURITY

PRINCIPLES & BEST PRACTICES

Network security refers to the policies, practices, and configurations designed to protect the integrity, confidentiality, and availability of data and resources as they travel across a network.

KEY PRINCIPLES



CONFIDENTIALITY

Ensure data is accessible only to authorized users.



INTEGRITY

Protect data from unauthorized alteration.



AVAILABILITY

Ensure systems and data are available when needed.



AUTHENTICATION

Verify the identity of users and devices.



ACCOUNTABILITY

Track and audit activities for responsibility.



BEST PRACTICES



Use strong passwords and multi-factor authentication.



Deploy firewalls and keep them updated.



Regularly update systems and patch vulnerabilities.



Be cautious of phishing and social engineering attacks.



Segment networks to limit lateral movement.



Monitor network traffic and analyze logs.



Backup data regularly and test recovery.

COMMON THREATS



MALWARE



DDoS
ATTACKS



PHISHING



MAN-IN-THE-
MIDDLE



UNAUTHORIZED
ACCESS



A secure network is built on a strong foundation of policies, tools, and awareness.



Secure today, protect tomorrow.

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CONCLUSION

YOU'RE NOW NETWORK-SMART!

You've completed the journey from the basics of networks to advanced concepts and security best practices. Keep learning, keep building, and stay curious—because the digital world is always evolving!



CONGRATULATIONS!

YOU'VE COMPLETED THIS TOPIC.
WELL DONE!



WHAT YOU'VE LEARNED



NETWORK FUNDAMENTALS

Understood what a network is, its components, types, and how devices communicate.



LAN & WLAN

Explored wired and wireless networks, topologies, devices, and real-world usage.



VPN & REMOTE ACCESS

Learned how VPNs work and how they provide secure access over the Internet.



THEORY & MODELS

Studied key theories and models like OSI Model and important network terms.



ROUTING & SWITCHING

Learned about IP addressing, routing protocols, and switching concepts.



NETWORK SECURITY

Understood security fundamentals, threats, and best practices to protect networks.



APPLICATIONS & PROTOCOLS

Explored application layer protocols and how data is exchanged.



PRACTICAL KNOWLEDGE

Gained practical insights to design, troubleshoot, and secure networks.

WHAT'S NEXT?



KEEP PRACTICING

Apply your knowledge with labs and real-world scenarios.



KEEP LEARNING

Explore advanced topics like CCNA, Cloud, Cyber Security, and more.



JOIN COMMUNITIES

Engage with peers, forums, and networking communities.



STAY CURIOUS

Technology evolves fast—keep exploring and stay ahead!



"THE NETWORK IS YOURS TO MASTER."
KEEP BUILDING. KEEP SECURING. KEEP GROWING.



THANK YOU!

Keep learning & stay awesome!

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NETWORKING TOPOLOGIES & CONNECTIONS



Understanding different network layouts
and the connections that bring the world together.

IN THIS TOPIC



NETWORKING
TOPOLOGIES



CONNECTIONS



EFFICIENT DESIGNS

Optimize network
performance.



STRONG CONNECTIONS

Reliable links for seamless
communication.



SECURE NETWORKS

Build safe, stable,
and trusted systems.



GLOBAL COMMUNICATION

Connect people, devices,
and organizations.

STAY CURIOUS. STAY SECURE.

WHAT ARE CONNECTIONS?

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Connections are communication mediums used to transfer data between network devices.



DATA SENT →



← DATA RECEIVED



SIMPLE EXPLANATION

Devices communicate using cables or wireless signals.



REAL LIFE EXAMPLES



HOME WI-FI

Wireless connection that connects all your home devices.



OFFICE LAN

Wired or wireless network in offices for sharing resources.



MOBILE INTERNET

Mobile data connection that keeps you connected anywhere.



FIBER BROADBAND

High-speed internet using fiber optic cables for fast communication.



CONNECTIVITY

Enables devices to talk to each other.



DATA EXCHANGE

Allows data, files, and messages to be shared.



BETTER PERFORMANCE

Strong connections improve speed and reliability.



ALWAYS CONNECTED

Keeps the world connected 24/7 across devices.

TYPES OF CONNECTIONS @manishburdak45

Connections can be classified into two main categories based on the medium used.



WIRED CONNECTIONS

Use physical cables to transmit data.



COAXIAL CABLE

Strong shielding with copper core.
Used in older networks and cable TV.



Cable TV



TWISTED PAIR CABLE

Most common LAN cable.
Used in homes, offices and computer labs.



LAN / Ethernet



FIBER OPTIC CABLE

Uses light signals for ultra-fast and long-distance communication.



Data Centers



**BOTH TECHNOLOGIES
HELP DEVICES COMMUNICATE
ACROSS NETWORKS.**



WIRELESS CONNECTIONS

Use radio waves or signals to transmit data.



WI-FI

Uses radio waves to connect devices to a network.
Common in homes and offices.



Home Wi-Fi



CELLULAR NETWORK

Mobile data connection through towers.
Used in 3G, 4G, 5G mobile networks.



Mobile Internet



SATELLITE CONNECTION

Uses satellites to provide internet in remote and difficult areas.



Remote Areas



Wired connections provide stability and speed, while wireless connections offer mobility and convenience. Both are essential in modern networking.



COAXIAL CABLE

AN OLDER WIRED COMMUNICATION CABLE
WITH STRONG SHIELDING.



DEFINITION

Coaxial cable is a type of copper cable that has a central conductor surrounded by insulation, a metallic shield, and an outer protective layer.



REAL LIFE USES

- Cable TV
- Old Ethernet Networks (10BASE2, 10BASE5)
- CCTV Systems
- Radio Frequency (RF) Communication



ADVANTAGES

- Better noise protection due to strong shielding.



DISADVANTAGES

- Slower than fiber optic cables.
- Bulky and less flexible.



Coaxial cables use a central copper core and shielding to reduce interference and provide stable communication over moderate distances.



SPEED
Moderate



DISTANCE
Medium

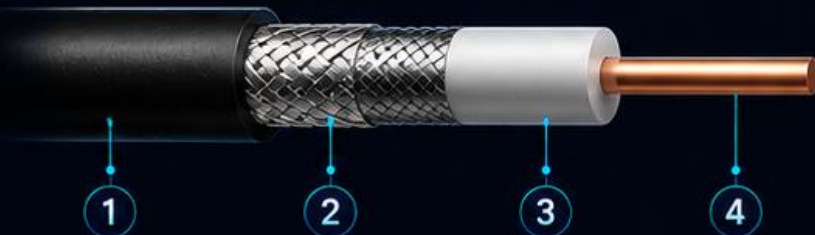


SECURITY
Moderate



COST
Low

CABLE STRUCTURE



1

OUTER JACKET
Protects the cable from damage.

2

BRAIDED SHIELD
Protects from electromagnetic interference (EMI).

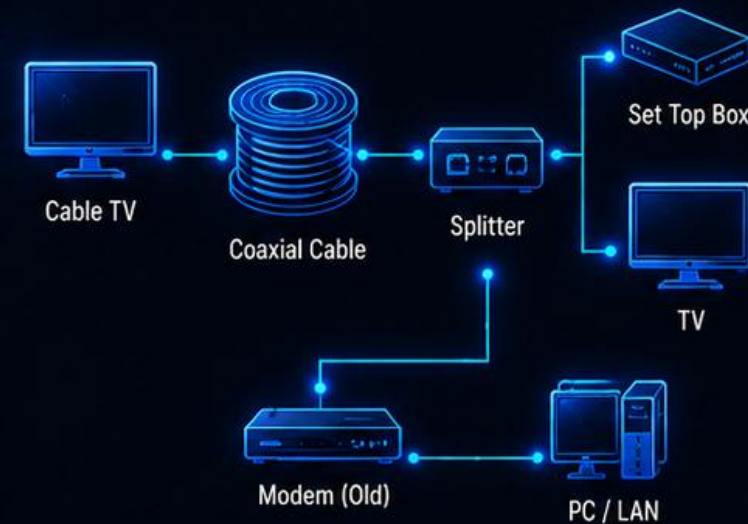
3

INSULATION
Separates the shield from the central conductor.

4

COPPER CORE
Carries the electrical signal.

REAL LIFE NETWORK EXAMPLE



KEY POINT

Coaxial cables were once the backbone of many communication systems and are still used in cable TV and CCTV installations.

TWISTED PAIR CABLE

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MOST COMMON NETWORKING CABLE
USED IN LAN CONNECTIONS.



DEFINITION

Twisted pair cable consists of pairs of insulated copper wires twisted together to reduce electromagnetic interference.



TYPES

- UTP (Unshielded Twisted Pair)
- STP (Shielded Twisted Pair)



REAL LIFE USES

- Home routers
- Office networks
- Computer labs
- IP phones
- Network switches connection

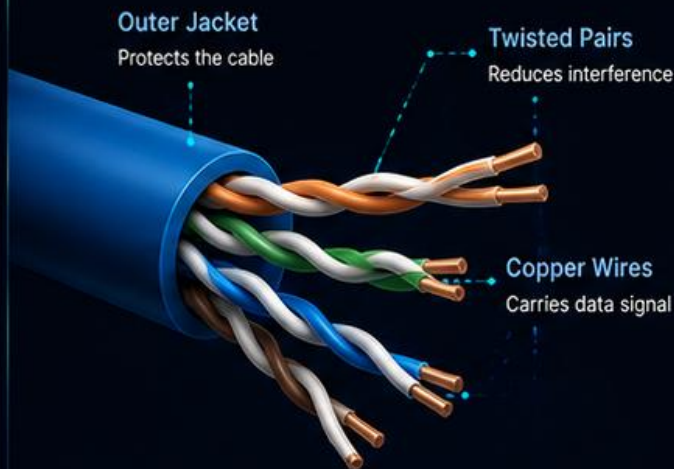


REAL LIFE EXAMPLE

An Ethernet cable connected from your laptop to a Wi-Fi router for internet access.



INTERNAL STRUCTURE



KEY POINT

Twisted pair cables are affordable, easy to install, widely used, and provide good speed for LANs.



ADVANTAGES

- Low cost
- Easy installation
- Flexible and widely available



DISADVANTAGES

- Affected by distance
- More interference than fiber
- Lower speed than fiber

SPEED COMPARISON

CABLE CATEGORY	MAX SPEED
Cat5e	1 Gbps
Cat6	10 Gbps
Cat6a	10 Gbps (more stable)
Cat7	10 Gbps+
Cat8	25/40 Gbps

LAN CONNECTION EXAMPLE



Twisted pair cable is the backbone of modern networking and is used in millions of devices every day.



AFFORDABLE

Low cost solution



RELIABLE

Stable connections



WIDELY USED

Used everywhere



EASY TO INSTALL

Plug and play

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FIBER OPTIC CABLE

USES LIGHT SIGNALS FOR ULTRA-FAST COMMUNICATION.



DEFINITION

Fiber optic cable uses thin strands of glass or plastic to transmit data using light pulses instead of electricity.



HOW IT WORKS

Light from a laser travels through the fiber core and reflects internally to reach the destination with very little loss.



CABLE STRUCTURE



Outer Jacket
Protects from damage

Kevlar Strength Member
Provides tensile strength

Cladding
Reflects light back into the core

Core
Carries light signals



FASTEST MODERN COMMUNICATION MEDIUM

High speed • Long distance • Low loss
• More secure • Future ready



DID YOU KNOW?

Fiber optic cables carry data using light, making them faster and more reliable than traditional copper cables.

ADVANTAGES



VERY HIGH SPEED

Supports Gbps to Tbps speeds easily.



LONG DISTANCE COMMUNICATION

Can transmit data over hundreds of kilometers.



LOW SIGNAL LOSS

Light signal weakens very slowly.



MORE SECURE

Difficult to tap or hack the data.



REAL LIFE USES



Jio Fiber



Airtel Fiber



Data Centers



ISP Backbone



FTTH
(Fiber to the Home)



SPEED

Up to 100x faster
than copper



DISTANCE

Up to 100+ km
without boosters



RELIABILITY

Works better in
harsh environments

WIRELESS CONNECTIONS

USE RADIO WAVES, CELLULAR NETWORKS, OR SATELLITES TO CONNECT DEVICES WITHOUT CABLES.



WI-FI (WLAN)

Uses radio waves to connect devices to a local network through a wireless router.



CELLULAR NETWORK

Uses mobile towers to provide internet access over wide areas to smartphones and other devices.



SATELLITE CONNECTION

Uses satellites orbiting Earth to provide internet in remote or hard-to-reach locations.



KEY FEATURES



MOBILITY

Connect from anywhere within range.



FLEXIBILITY

Easy to install and expand networks.



WIDE COVERAGE

Covers vast areas using cellular or satellites.



CONVENIENCE

No cables, more freedom to connect.

REAL LIFE EXAMPLES



Home Wi-Fi

Connecting phones, laptops, smart TVs, and IoT devices.



Mobile Hotspot

Sharing mobile internet with other devices.



5G Internet

High speed mobile internet for streaming, gaming, and more.



Remote Area Internet

Providing internet access in villages, deserts, and remote places.

COMMON USES



Online Streaming



Online Gaming



Video Calls & Meetings



Smart Home & IoT Devices



SECURITY TIP

Always use strong passwords, WPA2/WPA3 encryption, and keep your network updated to stay safe.



ENCRYPTION

Protects your data



AUTHENTICATION

Verifies users



FIREWALL

Blocks threats



UPDATES

Keep systems safe

WIRED VS WIRELESS

BOTH TECHNOLOGIES CONNECT DEVICES,
BUT EACH HAS ITS OWN STRENGTHS.

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WIRED CONNECTION



REAL LIFE EXAMPLE



- Gaming PC connected with Ethernet cable for low latency and stable connection.

FEATURE



WIRED



WIRELESS



SPEED

Generally Faster
High bandwidth

Slower than wired
(varies with signal)



SECURITY

More Secure
Harder to hack

Less Secure
Can be hacked



MOBILITY

Limited
Bound by cables

High
Move freely



STABILITY

Very Stable
Low interference

Affected by
interference



SETUP COST

Higher initial cost
(Cables & installation)

Lower initial cost
(Easy to setup)

WIRELESS CONNECTION



REAL LIFE EXAMPLE



- Mobile phone using Wi-Fi at home for internet access.



Wired networks are usually **faster**, **more secure**, and **more stable**, while wireless networks offer **mobility** and **convenience**.



Wired = Speed



Wired = Security



Wired = Stability



Wireless = Mobility



Wireless = Convenience

CYBER SECURITY IMPORTANCE

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SECURE CONNECTIONS PROTECT DATA,
PRIVACY, AND DIGITAL COMMUNICATION.



1. PACKET SNIFFING

Attackers capture data packets traveling over the network to steal sensitive information like passwords and logins.



2. WI-FI HACKING

Weak or unsecured Wi-Fi networks can be attacked by hackers to gain unauthorized access to devices and data.



3. WPA2 / WPA3 SECURITY

Use strong encryption protocols like WPA2 or WPA3 to secure wireless networks from unauthorized access.



4. SECURE NETWORK COMMUNICATION

Encrypted and secure communication ensures data integrity, confidentiality, and protection from cyber threats.



WHY IT MATTERS



Protects
Personal Data



Prevents
Unauthorized Access



Ensures Data
Integrity



Builds Trust in
Digital Services

BEST PRACTICES



Use strong passwords



Enable WPA3 encryption



Keep firmware updated



Use firewalls and intrusion detection



Monitor network activity

REAL LIFE EXAMPLES



Banks secure online transactions using encrypted connections.



E-commerce sites protect customer information with SSL/TLS.



Emails and messages are encrypted to prevent data leaks.



Educational institutions secure student data and online exams.



REMEMBER

A secure connection is the first line of defense in the digital world.
Stay aware, stay updated, stay secure!



ENCRYPT
Your Data



VERIFY
Connections



DETECT
Threats Early



PROTECT
Your Network

COMMON CYBER THREATS



MALWARE

Malicious software designed to damage, steal data, or disrupt systems.



PHISHING

Fraudulent attempts to steal sensitive information by disguising as trustworthy.



RANSOMWARE

Locks your data or system and demands payment to restore access.



PASSWORD ATTACKS

Brute force, dictionary, or credential stuffing attacks to guess passwords.



SOCIAL ENGINEERING

Manipulating people into revealing confidential information or performing risky actions.



DDoS ATTACKS

Overwhelming servers or networks with traffic to make them unavailable.



DID YOU KNOW?

- 95% of cybersecurity breaches are caused by human error.
- Phishing is the #1 attack vector used by cybercriminals.
- Ransomware attacks occur every 11 seconds globally.

THE GOAL: PROTECT WHAT MATTERS



DATA



PRIVACY



DEVICES



REPUTATION



Cybersecurity is everyone's responsibility.
Be smart. Be safe. Be secure.

QUICK SECURITY TIPS



Think before
you click



Don't share
sensitive info



Avoid public Wi-Fi
for sensitive tasks



Lock your
devices



Log out from
shared devices

BEST PROTECTION PRACTICES



Use Strong Passwords

Use long, unique passwords and
a password manager.

Enable Multi-Factor
Authentication (MFA)

Add an extra layer of security
to your accounts.



Keep Software Updated

Updates fix vulnerabilities and
protect against exploits.



Be Cautious with Emails

Don't click links or open attachments
from unknown senders.



Use Firewall & Antivirus

Keep firewall and antivirus enabled
and up to date.



Backup Regularly

Keep regular backups in a secure
location or cloud.

REAL LIFE EXAMPLES

SONY

2011: Sony was hit by a major cyber attack that
exposed millions of user accounts and data.

WannaCry

2017: WannaCry ransomware attack affected more
than 200,000 computers in 150+ countries.

facebook

2019: Data breach exposed the personal data of
540 million Facebook users.

NEXT TOPIC

NEXT PART

INTRODUCTION TO NETWORKING